

Solve the following system:

E-Vex

$$\begin{cases} \ln(x+1) - \ln y = 3 \ln 2 \\ x - 4y = 7 \end{cases}$$

$$\ln(x+1) - \ln y = 3 \ln 2$$

$$\ln\left(\frac{x+1}{y}\right) = \ln 8$$

$$e^{\left(\frac{x+1}{y}\right)} = e^{\ln 8}$$

$$\frac{x+1}{y} = 8 \Rightarrow x+1 = 8y$$

$$\underline{x = 8y - 1}$$

$$\text{then: } 8y - 4y = 7 + 1$$

$$4y = 8$$

$$\boxed{y = 2}$$

$$\text{then: } x - 4(2) = 7$$

$$\boxed{x = 15}$$

$$S: \{(15, 2)\}$$

E-Vex